

PocketSmart AI

Project Planning Phase

1. Project Planning Overview

PocketSmart AI is planned as a web-based GenAI recommendation system that helps users make budget-based decisions for home decor, party planning and jewelry.

2. Project Goals

- Develop a simple and user-friendly recommendation system.
- Provide recommendations based on user budget and preferences.
- Integrate Generative AI for personalized suggestions.
- Support home decor, party and jewelry planning.
- Provide fallback recommendations when AI services are unavailable.

3. Project Scope

Included:

- User registration and login
- Home decor planner
- Party planner
- Jewelry planner
- Budget-based recommendations
- AI-generated recommendations
- Recommendation results and estimated cost
- Responsive web interface

Not Included:

- Online payment
- Direct product purchasing
- Real-time inventory management
- Jewelry outfit image analysis

4. Technology Plan

- **Programming Language:** Python
- **Backend:** FastAPI

- **Frontend:** HTML, CSS, JavaScript, Jinja2
- **Database:** SQLite
- **AI Service:** Google Gemini API
- **Development Tool:** Visual Studio Code
- **Version Control:** GitHub

5. Development Phases

1. Requirement Analysis
2. Project Design
3. Project Planning
4. Backend Development
5. Frontend Development
6. AI Integration
7. Testing
8. Documentation and Demonstration

6. Project Timeline

Phase	Activity
Phase 1	Requirement collection
Phase 2	System and UI design
Phase 3	Project planning
Phase 4	Backend development
Phase 5	Frontend development
Phase 6	Gemini AI integration
Phase 7	Testing and debugging
Phase 8	Documentation and demonstration

7. Resource Planning

Human Resources:

- Student Developer
- Project Guide / Faculty

Software Resources:

- Python
- FastAPI
- Visual Studio Code
- Google Gemini API
- GitHub
- SQLite

Hardware Resources:

- Laptop or computer
- Internet connection

8. Risk Management

Risk	Mitigation
AI API unavailable	Use fallback recommendations
API response errors	Error handling and validation
Incorrect user input	Input validation
Database errors	Proper database operations
Development delays	Phase-wise development and testing

9. Expected Outcome

The final system will provide users with budget-aware recommendations for home decor, party planning and jewelry through a simple web interface powered by FastAPI and Generative AI.

10. Project Planning Summary

The project will be developed phase by phase, starting with requirements and design, followed by development, AI integration, testing, documentation and demonstration. This structured approach will help ensure that PocketSmart AI is completed within the planned timeline.